

ENERGY AND INFRASTRUCTURE HAZARD

Oil, Gas & Hazard Pipelines

Buried pipelines are usually quiet and out of sight, but a damaged high-pressure line can create a fast-moving fire, explosion, or chemical-release emergency.



1 Understand the factor

What it is - and why it matters

Oil, gas, and hazardous-material pipelines move natural gas, crude oil, refined fuels, carbon dioxide, and other products between production areas, terminals, refineries, utilities, and customers. Systems range from large interstate transmission lines to local distribution mains and service lines. Risk is shaped by product, pressure, pipe age and material, corrosion control, depth, valves, nearby excavation, slope movement, wildfire, and emergency shutoff capability. Public maps are useful for screening, but they intentionally do not show every line or the exact surveyed location.

How risk can reach a household

- A gas release can form a flammable cloud that ignites as a flash fire or explosion, with intense heat and blast pressure.
- Liquid products can spread through soil, drainage channels, groundwater, or waterways and may create fire, vapor, and cleanup hazards.
- Excavation, vehicle impact, corrosion, defective welds, earth movement, or construction activity can damage a line or coating.
- Emergency isolation may require shutting valves and depressurizing long sections, so product can continue escaping after a rupture.

What people often miss

- Pipeline markers identify a corridor, not the exact depth or centerline; the operator must locate the pipe before digging.
- The National Pipeline Mapping System focuses on transmission and hazardous-liquid lines and does not display every gathering, distribution, or service line.
- Some easements allow operator access, vegetation clearing, excavation, testing, and limits on structures or deep-rooted trees.
- Low-frequency events can have high consequences; distance alone is not enough without product, pressure, terrain, and shutoff information.

? Five checks before buying

1

Search the National Pipeline Mapping System and local utility records, then identify the operator, product, line type, and emergency number.

2

Review the recorded easement and survey to understand its width, access rights, building restrictions, and responsibility for restoration.

3

Ask the seller and local agencies about prior leaks, repairs, odor calls, emergency exercises, excavation damage, and nearby valves or regulator stations.

4

Determine whether the home, school, driveway, evacuation route, or waterway is downhill or downwind from the corridor.

5

Discuss fire, explosion, pollution, land-restoration, temporary-housing, and service-line coverage with the insurer before purchase.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 08

H Health impacts

- A major rupture can cause burns, blast injuries, smoke inhalation, toxic exposure, asphyxiation, and trauma during evacuation.
- A small chronic leak may create odors or indoor-vapor concerns, but proximity to an intact pipeline is not itself proof of exposure.
- Petroleum liquids and some transported chemicals can irritate skin, eyes, and lungs; health effects depend on the actual product and dose.
- Road closures, power or gas shutdowns, displacement, and fear of recurrence can affect medication access, sleep, and mental health.

P Property impacts

- Fire, explosion, and heat can destroy buildings, vehicles, utilities, landscaping, and nearby structures beyond the pipeline easement.
- Easements can limit additions, pools, walls, grading, tree planting, and future development even when the pipeline operates normally.
- Liquid releases can contaminate soil or groundwater and require excavation, vapor mitigation, water replacement, or long-term monitoring.
- Disclosure concerns, emergency history, visible equipment, and buyer perception may affect appraisal, financing, and resale.

I Insurance implications

- Homeowners policies may cover sudden fire or explosion damage, but pollution, land, earth movement, and gradual seepage exclusions can create gaps.
- Loss-of-use coverage generally requires a covered loss; precautionary evacuation without property damage may have limited reimbursement.
- Cleanup of contamination on the owner's land may be disputed when there is no covered direct physical loss or when pollution exclusions apply.
- Operator liability and federal compensation rules do not replace the need to understand personal property, auto, flood, and umbrella coverage.

Possible long-term health implications of buying nearby

- For most residents, the central long-term issue is preparedness for a rare acute event rather than continuous exposure from a properly operating buried line.
- Recurring alarms, excavation activity, vegetation clearing, and uncertainty can create chronic stress even when measurements show no exposure.
- A confirmed leak, contaminated well, persistent petroleum odor, or vapor-intrusion pathway changes the health question and requires environmental testing.
- Households with limited mobility, young children, older adults, pets, or medical equipment should plan how to leave quickly without driving across the corridor.

Evidence lens

Pipeline rupture, ignition, and contaminant-transport mechanisms are well established. Parcel-level risk requires line-specific information; public maps are screening tools and are intentionally approximate.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 08

V Verify the actual risk

- Contact the operator for emergency information and use 811 before every excavation, even for fences, trees, irrigation, or mailbox posts.
- Keep the easement, survey, operator contact, product information, and local evacuation instructions with property records.

P Protect the property

- Do not build, grade, park heavy equipment, burn, or plant deep-rooted vegetation in the easement without written approval.
- Protect drainage and slopes that cross the corridor and report exposed pipe, dead vegetation, bubbling water, vapor, or unusual stains.

M Monitor and respond

- Know warning signs such as hissing, roaring, petroleum odor, gas odor, mist, fire, soil blowing, or an unexplained patch of dead plants.
- If a leak is suspected, leave on foot upwind or crosswind, avoid switches and engines, call 911 and the operator from a safe location, and do not investigate.

I Prepare insurance records

- Ask for written explanations of pollution, explosion, fire, land, utility-service, and evacuation provisions.
- Maintain an inventory and adequate dwelling, contents, additional-living-expense, auto, and umbrella limits.

REAL-LIFE EXAMPLE

2010 San Bruno Natural Gas Pipeline Rupture

September 9, 2010

A Pacific Gas and Electric natural-gas transmission pipeline ruptured in San Bruno, California, and the escaping gas ignited. The National Transportation Safety Board investigated an explosion and fire that killed eight people, injured many others, destroyed 38 homes, and damaged additional properties.

Why it matters: The disaster showed how a high-pressure transmission-line failure in a residential neighborhood can escalate rapidly and why accurate records, integrity management, emergency isolation, public awareness, and evacuation are essential.

[Open official incident record - ntsb.gov >](https://www.nts.gov/incident-reports/2010-09-09-sbrn001)

Questions to resolve before making a decision

1. Search the National Pipeline Mapping System and local utility records, then identify the operator, product, line type, and emergency number.
2. Review the recorded easement and survey to understand its width, access rights, building restrictions, and responsibility for restoration.
3. Ask the seller and local agencies about prior leaks, repairs, odor calls, emergency exercises, excavation damage, and nearby valves or regulator stations.
4. Determine whether the home, school, driveway, evacuation route, or waterway is downhill or downwind from the corridor.

Official sources and mapping tools

[PHMSA National Pipeline Mapping System - nps.phmsa.dot.gov >](https://nps.phmsa.dot.gov)[PHMSA pipeline safety information - phmsa.dot.gov >](https://phmsa.dot.gov)[811 safe-digging service - call811.com >](https://call811.com)